VID – Complete Workspace Architecture (Detailed)

📘 **VID – Complete Workspace Architecture (Detailed)**
Workspaces = independent modules (micro-systems).
Features = capabilities inside each workspace.
YANTRA = intelligence layer across the system.

How to read this document

- **Tier 1 (Core)**: Mandatory for all schools. Foundation modules.
- **Tier 2 (Recommended)**: Widely used. Strongly improves efficiency.
- **Tier 3 (Conditional)**: Enabled only when the institution requires it.
- **Tier 4 (Platform)**: System-level control modules powering the VID ecosystem.

Tier 1: CORE Workspaces (Mandatory for All Schools)

> These form the foundation of any CBSE / ICSE / IGCSE institution ERP.

1) Academic Coordinator Workspace

Purpose

- Central academic configuration and control.

Primary stakeholders

- Academic Coordinator
- Principal / Admin (oversight)

Used for (key capabilities)

- Curriculum setup: classes, sections, subjects
- Academic calendar planning: terms, holidays, events, exam windows

- Subject and faculty mapping: subject-teacher allocation rules
- Academic policies and configurations: grading scheme, attendance rules, promotion criteria
- Activity and assessment planning: assessments, rubrics, co-curricular activities

Core data managed

- Class and section master
- Subject master and curriculum structure
- Academic year and term setup
- Policies and configuration tables

Typical workflows

- Create academic year → create classes/sections → assign subjects → map faculty → publish calendar

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2) Student Workspace

Purpose

- Complete student lifecycle interaction system.

Primary stakeholders

- Students
- Class teachers
- Academic office

Used for (key capabilities)

- Student profile and records: demographics, documents, IDs
- Attendance tracking (view): daily attendance summary and history
- Timetable access: class timetable, substitutions, room info
- Academic performance and report cards: marks view, grades, feedback
- Co-curricular tracking: participation, clubs, achievements
- Fee status and history (view): paid, pending dues, receipts

Core data managed

- Student master profile
- Attendance history (student view)
- Marks and grades (student view)
- Co-curricular participation log

Typical workflows

- Student checks timetable → views attendance → views marks/report card → downloads receipts

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3) Faculty Workspace

Purpose

- Teacher operational and academic interface.

Primary stakeholders

- Faculty
- Department heads

Used for (key capabilities)

- Teaching schedules: assigned periods and classes
- Course handling: lesson plan, topic coverage tracking
- Attendance marking: daily period attendance
- Student performance updates: marks entry, remarks, skill assessments
- Assignment and assessment handling: homework, tests, submissions, evaluation
- Communication with students: announcements, messages, class instructions

Core data managed

- Faculty timetable
- Attendance entries (teacher input)
- Assessment items and marks entries
- Class communications

Typical workflows

- Take attendance → assign homework → create assessment → evaluate → publish marks

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4) Examination Workspace

Purpose

- Exam and evaluation management system.

Primary stakeholders

- Exam Controller
- Faculty (marks entry)
- Principal (overview)

Used for (key capabilities)

- Exam scheduling: exam calendar, subject-wise schedules, time slots
- Marks entry: internal/external, practical/theory components
- Report card generation: templates, grade calculations, remarks
- Result publishing: release controls, visibility rules
- Performance analytics: class averages, subject analysis, toppers, trends

Core data managed

- Exam master and schedules
- Marks and grading rules
- Report card templates

Typical workflows

- Create exam → publish schedule → open marks entry window → validate → generate report cards → publish results

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5) Admission Officer Workspace

Purpose

- Admission pipeline management.

Primary stakeholders

- Admission Officer

- Front desk / enquiry team

Used for (key capabilities)

- Lead/enquiry tracking: walk-ins, calls, website leads
- Application management: forms, documents, status tracking
- Admission approvals: eligibility checks, interview results, final decision
- Student onboarding: convert admitted applicant to student profile

Core data managed

- Enquiry and lead records
- Applicant profiles and documents
- Admission status pipeline

Typical workflows

- Capture enquiry → follow-up → receive application → verify documents → approve → onboard as student

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6) Payment Administration Workspace

Purpose

- Fee and financial transactions management.

Primary stakeholders

- Accounts team
- Cashier
- Admin oversight

Used for (key capabilities)

- Fee structure setup: academic year fee heads, installments, concessions
- Fee collection: counter and online reconciliations
- Online payments: gateway integration, payment status tracking
- Receipts and invoices: auto-generation, reprints
- Pending dues tracking: defaulters list, reminders

Core data managed

- Fee structure and fee heads
- Student fee ledger
- Payment transactions and receipts

Typical workflows

- Configure fee structure → generate demands → collect payments → issue receipts → track dues

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7) HRMS Workspace

Purpose

- Staff and employee lifecycle management.

Primary stakeholders

- HR team
- Admin
- Employees (self-service)

Used for (key capabilities)

- Employee records: personal, job, documents
- Attendance and leave management: leave requests, approvals, attendance logs
- Payroll processing: salary structure, deductions, payslips
- Staff performance tracking: appraisals, feedback cycles

Core data managed

- Employee master
- Leave and attendance records
- Payroll and payslips

Typical workflows

- Onboard employee → manage attendance/leaves → run payroll → generate payslips → track performance

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8) Timetable Workspace

Purpose

- Scheduling engine for the institution.

Primary stakeholders

- Timetable Coordinator
- Faculty
- Students (consumers)

Used for (key capabilities)

- Class timetable generation: section-wise periods and rooms
- Faculty timetable: teacher allocations and workload
- Substitution management: absent teacher replacement, temporary allocations
- Period allocation: subject-period rules, lab scheduling constraints

Core data managed

- Timetable master (class + faculty)
- Substitution logs
- Room/lab resources (if enabled)

Typical workflows

- Define constraints → generate timetable → publish → handle substitutions daily

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9) Common Workspace (Student Central)

Purpose

- Shared services across all users.

Primary stakeholders

- Students
- Faculty
- Admin staff
- Parents (if enabled)

Used for (key capabilities)

- Notice board: circulars and announcements
- Requests and approvals: certificates, leave requests, permissions
- Basic communication: messaging, broadcast updates
- Shared announcements: events, schedules, urgent alerts

Core data managed

- Notices and circulars
- Request tickets and approval status
- Communication logs

Typical workflows

- Publish notice → users acknowledge → create request → route to approver → close request

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10) Principal / Admin Dashboard Workspace

Purpose

- Strategic decision-making interface.

Primary stakeholders

- Principal
- Management
- Admin leadership

Used for (key capabilities)

- Institution-level analytics: attendance trends, fee collections, academic progress
- Reports overview: consolidated reports across workspaces
- Monitoring operations: compliance, pending approvals, exceptions
- Performance insights: teacher performance indicators, student outcomes, operational KPIs

Core data managed

- Dashboard metrics (aggregated)
- Summary reports and drill-down links

Typical workflows

- Review daily dashboard → identify exceptions → drill into relevant workspace → take action / assign follow-ups

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Tier 2: RECOMMENDED Workspaces (Widely Used)

> These significantly improve operational efficiency.

11) Transport Coordinator Workspace

Purpose

- Transport system management.

Primary stakeholders

- Transport Coordinator
- Drivers / attendants (if enabled)

Used for (key capabilities)

- Bus routes and stops: route planning, stop master
- Student allocation: assign students to route/stop
- Transport tracking: bus status, trip logs (if GPS enabled)
- Route optimization: capacity planning and route efficiency

Core data managed

- Vehicle and route masters
- Stop master
- Allocation lists

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12) Library Management Workspace

Purpose

- Digital library system.

Primary stakeholders

- Librarian
- Students and faculty

Used for (key capabilities)

- Book inventory: catalog, copies, categories
- Issue/return tracking: due dates, renewals, fines
- Usage monitoring: popular books, borrower history

Core data managed

- Book and copy master
- Issue/return logs
- Fine ledger (optional)

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13) Parent Portal Workspace

Purpose

- Parent engagement system.

Primary stakeholders

- Parents/guardians
- Class teachers
- Admin

Used for (key capabilities)

- Student progress tracking: marks, grades, remarks
- Attendance visibility: daily and monthly summaries
- Fee payments: pay online, download receipts
- Communication with school: messages, notices, meeting requests

Core data managed

- Parent-linked student views
- Communication and acknowledgements
- Payment status views

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14) Event Management Workspace

Purpose

- School events and activities management.

Primary stakeholders

- Event Coordinator
- Activity clubs
- Students and faculty

Used for (key capabilities)

- Event planning: event master, agenda, budgets (optional)
- Participation tracking: registrations, attendance, winners
- Activity scheduling: calendar alignment and resource booking

Core data managed

- Events master
- Participant lists
- Schedules

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15) Disciplinary Committee Workspace

Purpose

- Student behavior and discipline tracking.

Primary stakeholders

- Discipline Committee
- Class teachers
- Principal (oversight)

Used for (key capabilities)

- Incident reporting: date/time, description, evidence
- Action tracking: counselling, warnings, parent meetings
- Behavior records: student-wise discipline history and patterns

Core data managed

- Incident log
- Actions and follow-ups

- Student behavior profile

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16) Sports Officer Workspace

Purpose

- Sports and extracurricular management.

Primary stakeholders

- Sports Officer
- Coaches
- Students

Used for (key capabilities)

- Sports events: intra/inter-school events scheduling
- Team management: selections, squads, roles
- Performance tracking: results, records, achievements

Core data managed

- Teams and players
- Match/event schedule
- Performance stats

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17) Health Monitoring Workspace

Purpose

- Basic student health tracking.

Primary stakeholders

- School nurse / health coordinator
- Admin

Used for (key capabilities)

- Medical records: allergies, conditions, vaccinations
- Emergency logs: incidents and treatments
- Health alerts: flagged cases, special instructions

Core data managed

- Student health profiles
- Incident and emergency log

Tier 3: CONDITIONAL Workspaces (Institution-Specific)

> Activated based on institution type.

18) Hostel Administration Workspace

Purpose

- Hostel allocation and operations management.

Primary stakeholders

- Hostel Warden
- Admin

Used for (key capabilities)

- Hostel allocation: rooms, beds, assignments
- Hostel attendance: roll call, night attendance
- Gate passes: approvals, entry/exit logs
- Hostel operations: complaints, maintenance, mess (optional)

Core data managed

- Hostel room/bed master
- Allocation registers
- Gate pass logs

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19) Alumni Management Workspace

Purpose

- Alumni database and engagement system.

Primary stakeholders

- Alumni Coordinator

- Management

Used for (key capabilities)

- Alumni database: batch-wise alumni records
- Engagement and communication: newsletters, updates, invites
- Networking: mentorship, alumni contributions

Core data managed

- Alumni master
- Engagement history

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20) Placement Cell Workspace

Purpose

- Job placement management (mostly for colleges).

Primary stakeholders

- Placement Officer
- Students
- Employers (external)

Used for (key capabilities)

- Job placements: drives, job postings, eligibility
- Company interactions: MoUs, schedules, interview rounds
- Student career tracking: applications, offers, outcomes

Core data managed

- Company master
- Job/drive records
- Student application pipeline

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21) Research & Development Workspace

Purpose

- Higher education research management.

Primary stakeholders

- Research cell
- Faculty researchers

Used for (key capabilities)

- Research projects: proposals, milestones, funding (optional)
- Publications: journals, conferences, indexing
- Innovation tracking: patents, prototypes, startups (optional)

Core data managed

- Projects register
- Publications register
- Innovation register

Tier 4: PLATFORM Workspaces (System-Level Control)

> These power the VID ecosystem itself.

22) Team Owner Workspace (Super Admin)

Purpose

- Global platform control across institutions.

Primary stakeholders

- Platform owner / super admin

Used for (key capabilities)

- Institution creation and management: tenant setup, activation/deactivation
- Subscription and billing control: plans, payments, renewals
- Global analytics: adoption, usage, performance metrics
- System monitoring: uptime, errors, service health

Core data managed

- Institution registry

- Subscription and billing records
- Platform-level metrics

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23) User Management System Workspace

Purpose

- Identity and access control.

Primary stakeholders

- Admin
- Platform security team

Used for (key capabilities)

- User creation: students, parents, faculty, staff
- Role assignment: role mapping per institution and module
- Permission control (RBAC): what each role can view/edit/approve

Core data managed

- Users directory
- Roles, permissions, policies
- Audit logs (optional)

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24) Course Coordinator Workspace

Purpose

- Course-level structuring (advanced).

Primary stakeholders

- Course Coordinator
- Academic leadership

Used for (key capabilities)

- Course definitions: courses, streams, electives
- Course allocation: sections/batches, faculty assignment
- Curriculum mapping: outcomes, modules, skill mapping

Core data managed

- Course catalog
- Allocation matrices
- Curriculum map

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25) YANTRA Workspace (AI Engine)

Purpose

- Intelligent automation layer across VID.

Core modules inside YANTRA

- Voice Agent
- Chatbot
- Face Recognition Attendance

Used for (key capabilities)

- AI-based student/parent interaction (FAQ, guidance, support)
- Voice-enabled admin operations (quick actions, status checks)
- Automated support system (ticket triage, knowledge base responses)
- Smart attendance (face recognition workflows where applicable)

Cross-functional nature

- YANTRA is not a standalone school department module. It enhances and automates actions across all workspaces.

Final architecture summary

VID Workspace Ecosystem
CORE (10): Academic, Student, Faculty, Examination, Admission, Payments, HRMS, Timetable, Common, Principal Dashboard
RECOMMENDED (7): Transport, Library, Parent Portal, Events, Discipline, Sports, Health
CONDITIONAL (4): Hostel, Alumni, Placement, R&D
PLATFORM (4): Team Owner, User Management, Course Coordinator, YANTRA (AI)

Key understanding

- Workspaces are independent modules.
- Features are implemented inside each workspace.
- YANTRA is the intelligence layer across the full system.

Notes for implementation (optional, but useful)

- Each workspace should have clear **roles**, **permissions**, **data ownership**, and **audit trail**.
- Any cross-module feature should be implemented through controlled integrations and shared identifiers (Student ID, Employee ID, Academic Year, Class-Section).
- Dashboards should pull from source workspaces, not duplicate data.